

Rupesh Kumar

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Skills

Languages: Python, Golang, SQL, GraphQL

Backend & APIs: REST APIs, gRPC, Apollo Federation GraphQL, Microservices, Event-Driven Architecture

Databases & Storage: MySQL, PostgreSQL, Redis, Vertica, Apache Iceberg

Messaging & Streaming: RabbitMQ, Kafka

Cloud & Infrastructure: AWS (EC2, S3, SQS, Lambda, Athena, RDS), Docker

Observability & Monitoring: Prometheus, Grafana, OpenTelemetry, Kibana, CloudWatch, New Relic, Dynatrace

Distributed Systems: High Availability (HA), Disaster Recovery (DR), Data Synchronization, Virtual IP (VIP)

Education

IIIT Dharwad

Aug 2018 - Jun 2022

B.Tech in Computer Science and Engineering

Work Experience

Sandvine Technologies (Applogic Networks), Bangalore

March 2025 - July 2026

Software Development Engineer - 2

- Designed and implemented active-passive **High Availability (HA)**, **Virtual IP (VIP)** failover, disaster recovery, and cross-node data sync mechanisms; developed **confd** integrated CLI tools for data reconciliation, hourly/minutely-granularity data synchronization, and synchronization progress monitoring across distributed deployments.
- Developed Python-based Correlation Engine pipelines to process **VoIP connection data** through normalization, single-sided call detection, correlation, and **Call Detail Record (CDR)** generation workflows; optimized normalization and SQL processing, reducing execution time from **10 to 6 minutes** for **100K records/minute** across 15-minute data windows, with centralized tracking of pipeline execution status and processing windows.
- Built **Golang-based microservices and APIs** to serve correlated call records and support filtered data retrieval based on customer-defined criteria; Implemented **on-demand historical record generation** workflows for data unavailable in precomputed datasets.
- Developed a **Pytest-based test automation framework** for IDS (a Vertica-based analytics platform), automating validation of customer-specific features, analytics workflows, and database functionality through **integration and regression testing**.

Sling Media, Bangalore

July 2022 - March 2025

Software Development Engineer

- Enhanced Sling TV's shared metadata server performance by implementing an optimized filtering mechanism using Golang, GraphQL, and Consul — reducing latency and server load for Video Ribbons and Tiles, and ensuring a seamless experience for **2M+** subscribers during **high-traffic VOD operations**.
- Designed and built modular image metadata services (Ribbon, Tiles, EPG) using Golang, gRPC, and Apollo Federation GraphQL — powering real-time content discovery for **Sling's OTT** platform with 750ms GET latency (95% SLO) and <4% error rates (100% SLO).
- Optimized video management workflows by decoupling media clip and processing flows (Flask, Django, RabbitMQ), containerizing with Docker — reduced deployment time by 30% and accelerated background task processing.
- Implemented a scalable channel health monitoring system aggregating multi-source stream metadata, computing segment anomalies, and storing results efficiently in S3 using Apache Iceberg, enabling real-time diagnostics and proactive issue detection via Pylceberg.
- Technologies: Golang, Python, GraphQL, Apollo Federation, gRPC, Kafka, RabbitMQ, Docker, AWS (EC2, S3, Lambda), MySQL, Redis, Elasticsearch, Consul.
- Observability: Prometheus, Grafana, OpenTelemetry, Kibana, AWS CloudWatch.

Shipx, Bangalore

Jan 2022 - July 2022

Software Development Engineer Intern

- Implemented indexing and partitioning on a MySQL database to optimize query performance, prepared data using SQL for Metabase integration, and embedded reports into a Ruby on Rails web module.
- SQL, HTTP, REST API's, Ruby-on-Rails.